

Context and Problem Statement:

The core problem of the SLF is the fact that they are landing on grape vines and infesting grapes. Once the grapes get infested, there are currently no effective methods to separate SLF from harvested grapes. Our objective is to prevent the issue before it becomes one by preventing the SLF from reaching the vines in the first place. To tackle this, we opted for a fly trap that uses light as a lure, which allows us to avoid chemicals such as pesticides that are harmful to grapes. The idea was to place these traps outside of grapefields at night to attract and trap SLF, preventing them from getting to the grapes.



Fig 1: Constructed Final Prototype

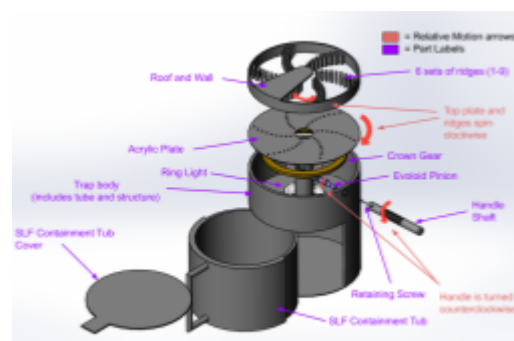


Fig 2: Final Prototype exploded view. Legend at top right.

Final Prototype and Application:

The device we have fabricated is a light trap that lures SLF away from the grape vines and traps them until they can be disposed of. The main mechanism of the trap is a clear, spinning plate with ridges that allow light to pass through, attracting SLF. The device works by allowing SLF to land on the plate and forcing them to either move into the tube at the center of the trap or be crushed by the interlocking ridges. The trapped bugs will be stored in the bottom container of our device and can be emptied at the user's convenience. The user can then clean out the SLF from the container and either

throw them away or reuse them as feed for animals.

Assembly Process:

Only two batches of assembly need to take place to build our device. After all the parts are manufactured, you first insert the pinion gear through the hole in the main body and into the crank, then secure it with a through bolt, as described in Figure 3. Next, the ridges press-fit into the holes on the top side of the acrylic plate in descending size order, with the largest ridges on the outside. These were secured with super glue to ensure rigidity. Then, the crown gear is superglued to the bottom side of the acrylic plate, ensuring it is centered. The 3D printed roof is then secured with superglue to the top of the tube, which is part of the device body, and can be viewed in Figure 4. Lastly, the containment tub and cover can be slid into the device body. A fully exploded CAD assembly is shown in Figure 2.

Conclusion and Recommendation

Overall, our device met all our success criteria and hit a relatively low cost of \$118.75. Quantifiable results supporting this decision can be seen in the testing section. In addition, the parts are easily fabricable, and the device is easy to assemble. For these

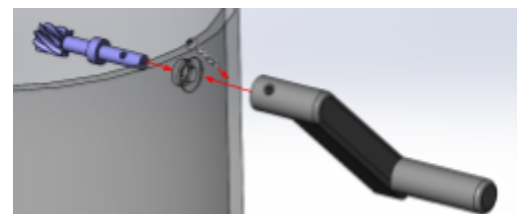


Fig 3: Crank mechanism is assembled by pushing the helical pinion through the wall of the device and into the crank. A bolt then goes through both to secure.

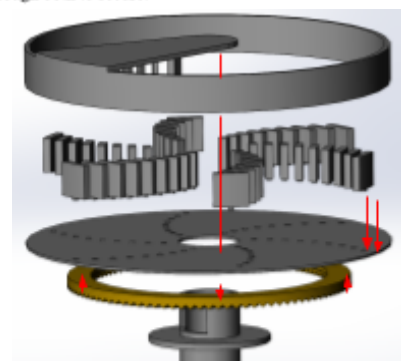


Fig 4: The top part of the device is assembled by glueing the crown gear to the acrylic plate, then glueing the ridges into their holes, and then locking everything in by glueing the roof to the tube.

reasons, we recommend moving forward with this prototype by making slight modifications and performing real-world testing.

The first modification we would need to make is to replace the crank with a motor for ease and efficiency of use. This obvious modification would also enable our device to operate without human supervision. Then, we would replace our 3D printed gear and pinion setup with a metal one. 3D printed gears wear down quickly, but we were forced to use them since McMaster didn't have any crown gears big enough to fit around the shaft of our device. We would recommend finding a large gear to purchase or outsourcing a custom gear to fit around the shaft and withstand the wear of the motor.

We would also have to make modifications to accommodate real-world operating conditions. Our only client feedback was that our device needed to be placed away from the grape fields so that it didn't unintentionally lure SLF to uninfested fields. Because of this necessary distance, we would recommend looking into solar power as a method for powering both the light and the motor. Since our device is primarily meant to operate at night, solar power could be accumulated during the daylight hours and then discharged through the evening. Finally, our device requires routine cleaning to remove SLF from the trap and clean dead bugs from around the ridges. Serviceability modifications may need to occur depending on how the device holds up in a real-world environment.

In terms of further testing, the main thing we would need to look into is the specific type of light that attracts SLF. There hasn't been a lot of research on SLF being attracted to light, and we would want to ensure that we are attracting as many SLF as possible to the device. We would also want to test our device in the field to make sure that nothing goes wrong when it is faced with actual bugs instead of paper towel mockups. However, due to our initial success, we think that moving on to this phase of testing would be a good next step.

Testing and Results:

In order to ensure that our prototype met our success criteria, we designed tests that would provide an accurate measure of its effectiveness. Firstly, we wanted our prototype to have a large capacity for SLF so it would not have to be serviced as often. We set a target capacity of 1000 SLF as a large overestimate of the number of SLF that would enter the trap per cycle. Indeed, according to a report by Heather and Ashley Leach [1], an average of around 50 SLF were observed entering the target vineyards per day during peak season, so our device would be able to go at least a week without service. We tested this by ensuring that the bottom drawer would open and close consistently and by ensuring that its volume was great enough to hold the SLF. We estimated that we would need a minimum of 4 liters of storage space to hold 1000 SLF, and we checked in CAD to ensure that we achieved this volume. Given that this is a smaller model of what the final product could be, a larger version could potentially go multiple weeks without needing cleaning, making it easier for grape farmers to manage.



Fig 5: Testing light emission

The second success criterion we established was that our trap should emit and disperse light effectively. Specifically, we wanted at least 80% of the light from our light source to pass through the trap top in order to minimize power demands and ensure that the strong light didn't end up overheating the trap. We tested this by measuring the lux coming from our ring light both inside and outside of our trap with a phone app that utilized the phone's light sensor, as seen in Figure 5. We held the phone 5 inches above the light and used a piece of paper as a diffuser. We found that the light dropped very little, from

14400 to 14200 lux, and determined that 98.6% of the light was able to pass through the trap. Our design will be able to let through sufficient light to lure the SLF without expending massive amounts of power.

Our third success criterion was that our trap could handle multiple SLF landing on a section of it at once, as could occur in actual use. We did not have sufficient data on how SLF are attracted to light, so we set a target of handling 3 SLF landing on the trap at once, which we estimated was a reasonable figure. We made small SLF models out of paper towels and tape and attempted to push them through the trap. On our first prototype, where it was easier to simulate the motion of the SLF by pushing them, we were able to fit 6 SLF models through the central tube at once without any getting stuck. On our final, we were able to shake the trap to simulate SLF crawling through the central tube, and 3 SLF were able to be shaken into the holding container at the bottom. This indicates that our trap will be able to effectively handle trapping multiple flies at once.

Our final success criteria was that the trap would not allow SLF to escape once they were on the plate, forcing them to either be crushed or to travel down the central tube. We tested this by placing our SLF models on top and rotating them to see if the flies would be able to escape. When the paper towel flies got trapped between the upper and lower ridges, there were no gaps for them to crawl through, and they were caught between the ridges of the trap, indicating that the bugs would be crushed once our design was constructed with stronger materials. This ensures that the SLF attracted to our trap will be unable to escape and will be forced to either travel into the trap's center or perish.

We also ran some basic structural tests to simulate real-world conditions such as inclement weather. We shook the machine at 5 Hz, spun the top at 7.25 rad/s, and threw our mock SLF at it to simulate flies landing on the top. It held up through all of those tests, implying that it will not need service often.

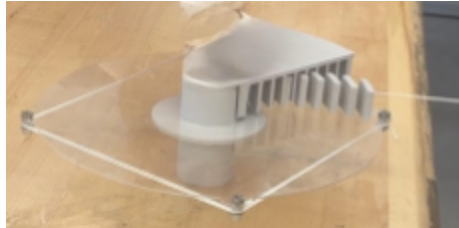


Fig 6: String being pulled to spin

Prototype and Testing Details:

Prior to designing and fabricating our final prototype, we built an initial prototype to confirm the functionality of our main mechanism, the spinning top. The main functionality we were hoping to confirm was that the acrylic lid ridges and roof ridges didn't collide, and that a threat of death was established by the ridges, which would force the SLF down the tube.

This prototype consisted of the upper components of our final device – the tube, acrylic plate, ridges, and roof. A string could then be wrapped around screws placed at the edge of the plate, and then pulled, to spin the device. By testing the spinning functionality of the acrylic plate, we were confident that when adding more sets of ridges on the final prototype, everything would work as intended. This testing is shown in Figure 6. Additionally, we tested to make sure that the threat of death was established by the ridges, while also forcing SLF down the tube, by using replica SLF made from paper towels. This testing is shown in Figure 7. As everything worked as planned, we decided to move forward with our design, with only small changes to its tolerances.

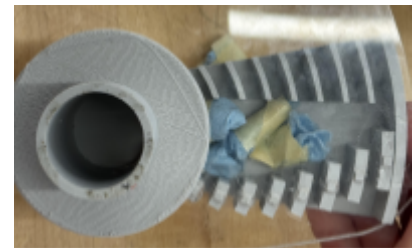


Fig 7: Testing threat of death

Reference

[1] – <https://link.springer.com/article/10.1007/s10340-020-01233-7>

BOM/Component List:

BOM of Initial Prototype

Part	Price
3D Prints	\$5.36
Acrylic	\$6.62
Laser Cutting	\$6.28
Total	\$18.26

BOM of Final Prototype

Part	Price
Light	\$19.99
3D Prints	\$65.26
Acrylic	\$6.62
Wood	\$1.75
Laser Cutting	\$25.13
Total	\$118.75

Total Spent	\$137.01
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Components:

3D-Prints:

- Ridges (x54 for final; x9 for prototype)
 - Specs: 0.875" tall, widths vary from .25"-1.25"
 - 3D printed out of PLA and done at the RPL
- Roof and Wall
 - Specs: radius = 4.5", matching ridges 3D printed with roof as one piece
 - 3D printed out of PLA and done at the RPL
- Crown and pinion
 - Specs: 96 and 8 teeth, respectively
 - 3D printed out of PLA and done at the RPL
- Crank
 - Specs: 4" long
 - 3D printed out of PLA and done at the RPL
- Trap Body
 - 9" diameter, 9.5" height
 - 3D printed out of PLA and done at the RPL
- Containment Tub
 - Specs: 295.3 in³ volume
 - 3D printed out of PLA and done at the RPL

Acrylic

- Clear Cast Acrylic Sheet (1 for prototype and 1 for final)
 - Specs: 12" x 12" x 1/16"; McMaster 8560K171
 - Laser cut at RPL

Wood

- 1/16" plywood 12"x12" sheet
 - Laser cut at RPL

Bolts and Nuts

- x4 each for initial prototype
- Free from TDS

Lantern-Lure

What if solving the SLF invasion was as simple as turning on a light?

Ashlyn Roeder, Asher Ciardiello, Ira Geller, Nicholas Letendre, Mamadou Barry

Preventing SLF Migration

- SLFs are using trees of heaven as a jumping-off point to reach grape fields
- SLFs are landing on the grapevines themselves, contaminating the grapes
- This issue of SLFs migrating to grape fields is detrimental to the economic feasibility of the grape industry

Initial Prototyping

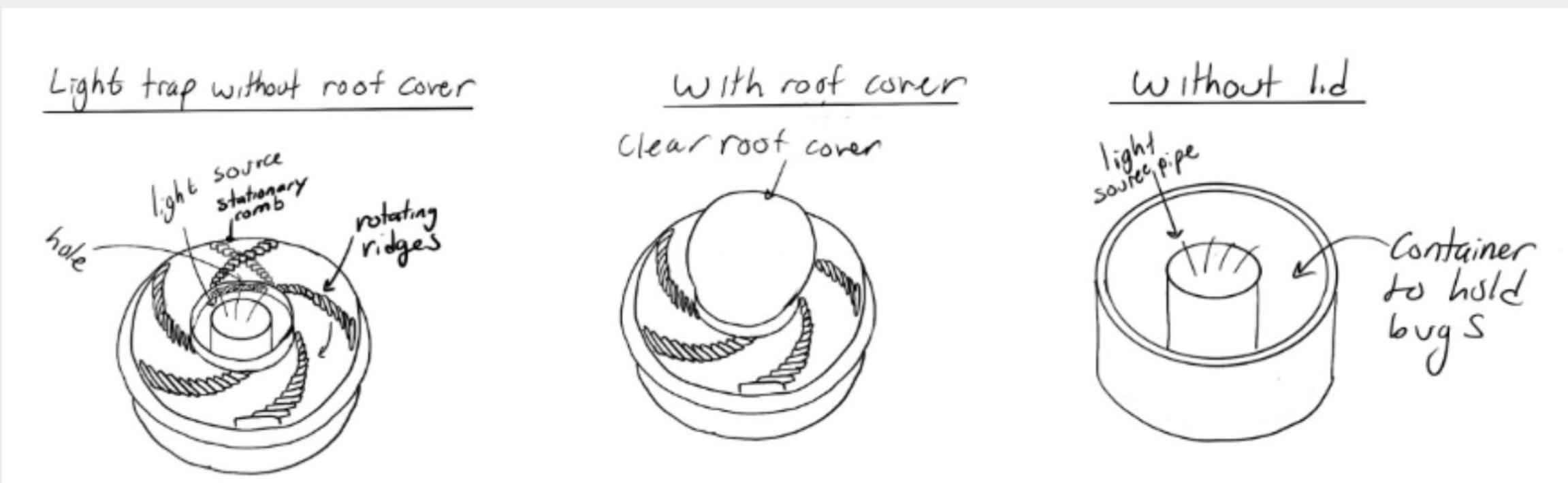


Fig. 1: Initial prototype sketches. Various views.

- **Spinning trap – either squishes or traps SLFs**
 - Avoids pesticides
 - Draws SLFs away from grape fields
- **Clients suggested keep away from fields**
 - Don’t want to bring SLFs to uninfested fields

Design Estimates from Cardboard Mockup

Light Power	40-100W
Total Electrical Power	120-250W
Daily Energy Use	1.2-2.5kWh
Motor Power	90-120W
Chamber Volume	4-12 Liters
Overall Trap Dimensions	r =14cm; h =24cm

Tab. 1: Design estimates. Ranges to meet goals.

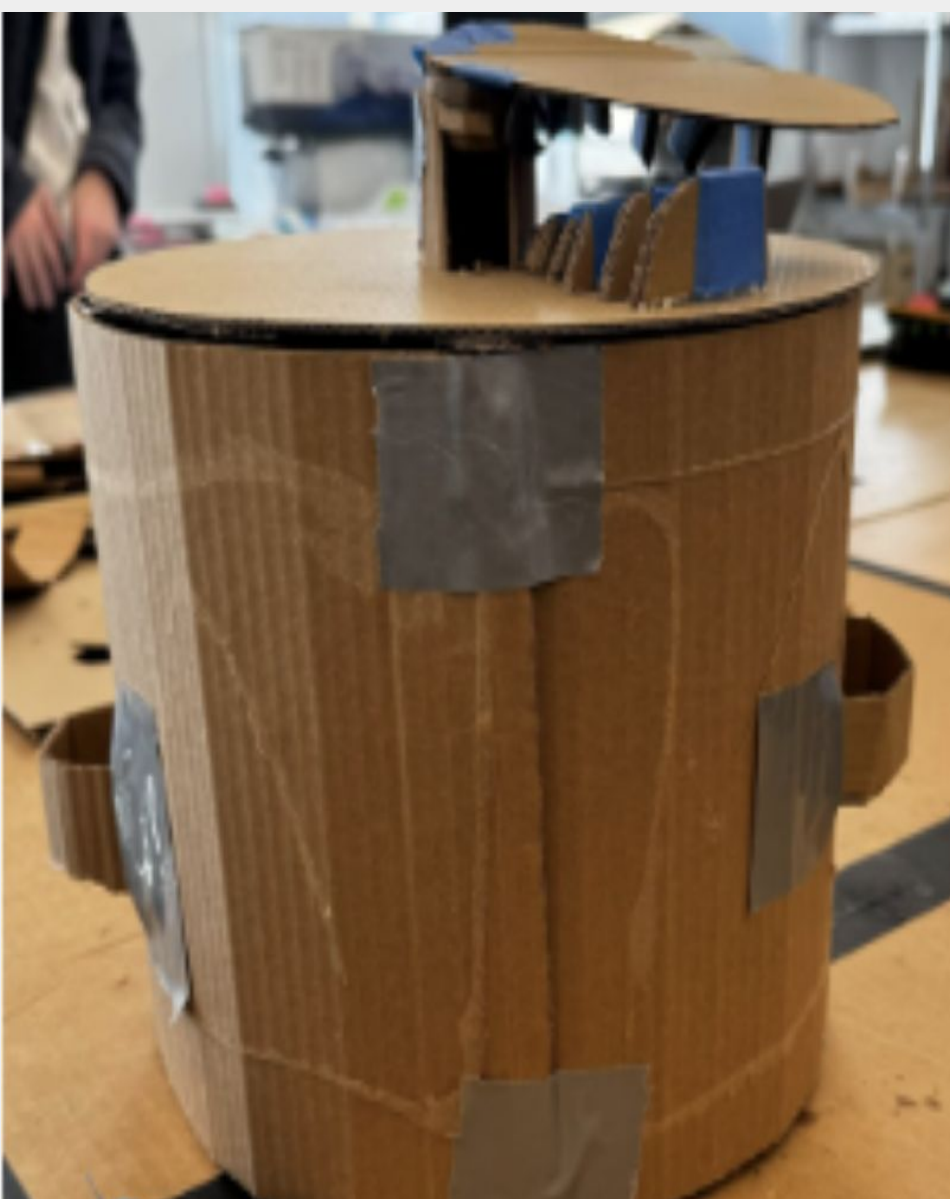


Fig. 2: Cardboard Mockup. Testing dimensions.

Main Mechanism Intial Testing

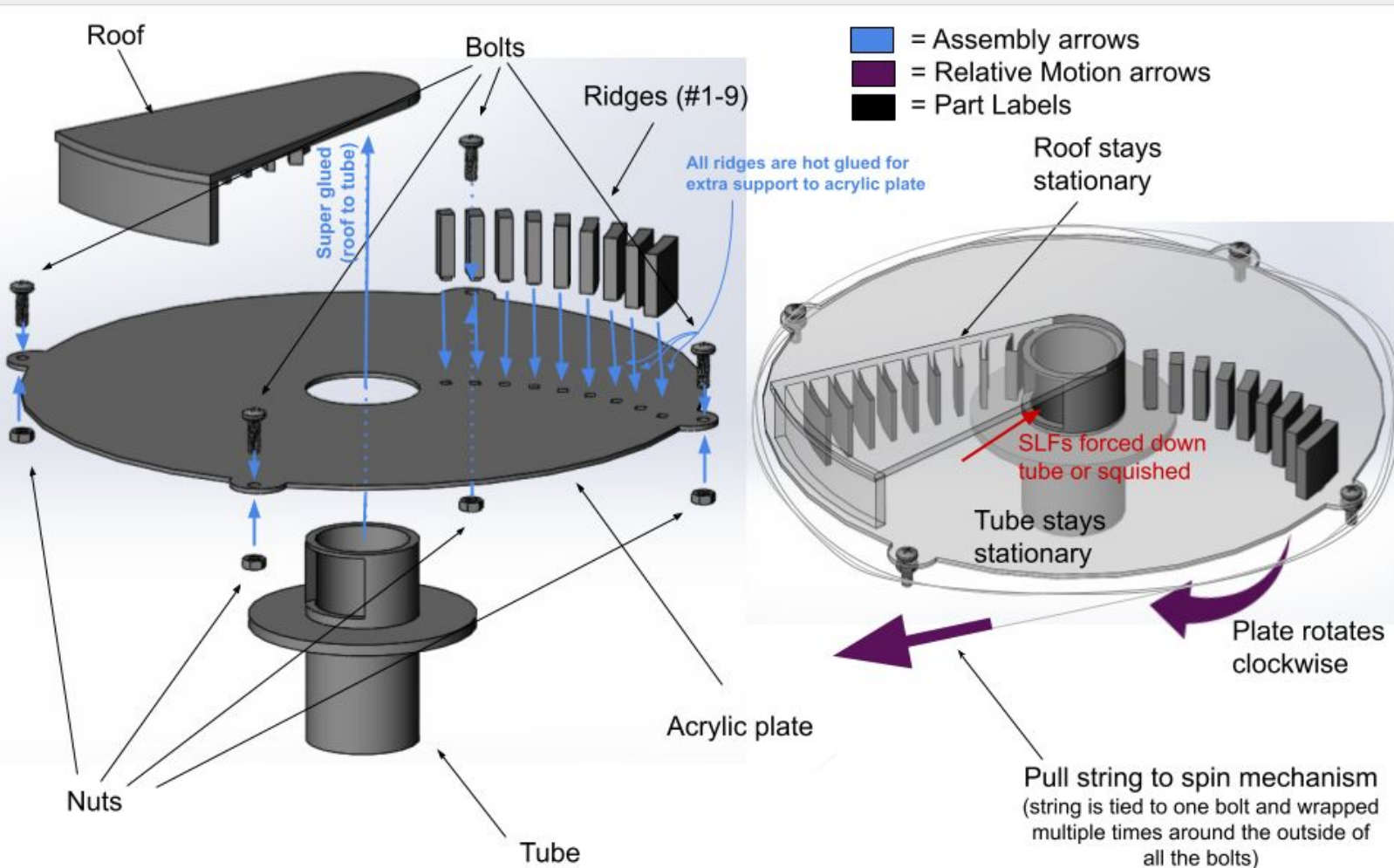


Fig. 3: Main mechanism testing. Exploded view and functionality diagram.

Goals:

- Test main mechanism (rotating top)
- Prove it traps and pushes SLFs down the tube

Success Criteria

- Sufficient light passes through top (≥80%)
- Sufficient capacity for dead SLF (≥1000)
- Can handle multiple flies getting trapped at once (≥3)
- Establishes threat of death for SLF (1=sucess, 0=fail)

Final Prototype

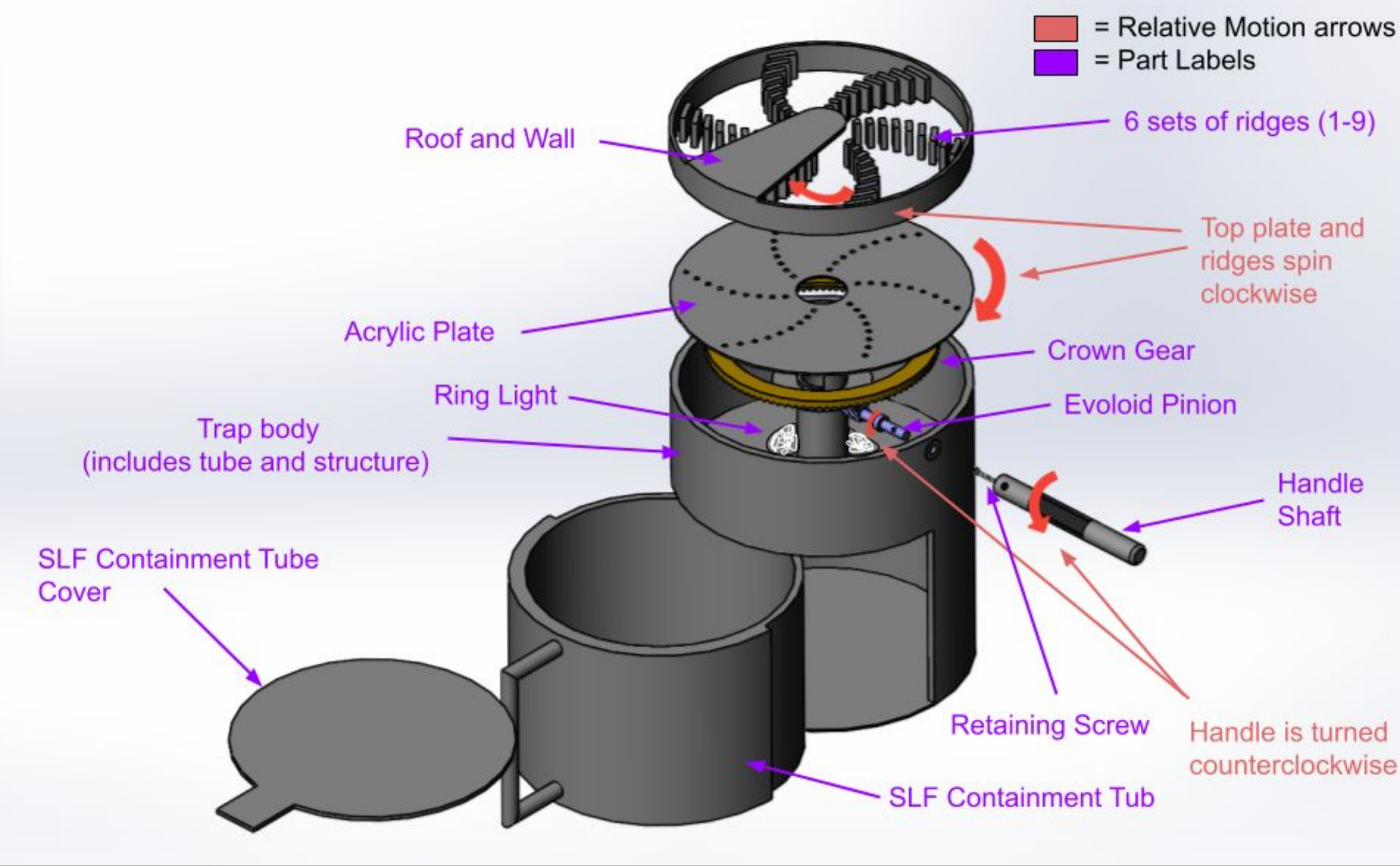


Fig. 4: Final Prototype exploded view. Legend at top right.

BOM

Light	\$19.99
3D Prints	\$70.62
Acrylic	\$13.24
Wood	\$1.75
Total	\$105.60

Tab. 2: Bill of Materials. Excludes costs of services (\$31.41) for laser cutting.

Components

- All components are listed above in purple
- SLF cover made from wood
- Plate from acrylic
- All else is 3D printed

Testing the Light, Functionality, and Capacity

- **Squish Test**
 - No gaps larger than ½” between ridges
- **Tube Capacity Test**
 - Can fit 6 SLF through tube

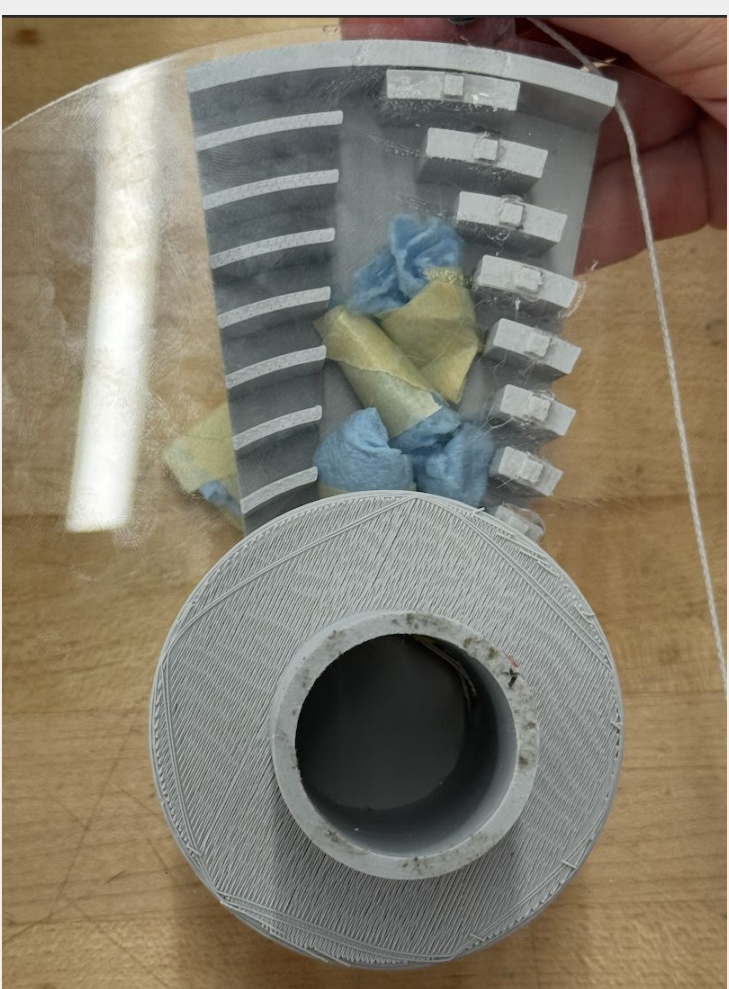


Fig. 5: Squish test. View below first prototype.



Fig. 6: Light test. Light emission in dark closet.

- **Light Test**
 - Dropped from 14,400 lux to 14,200 lux with acrylic
 - 98.6% of light passes through
- **Structural Tests**
 - Held up under simulated real world conditions

- All tests were successful – no changes needed
 - Due to improvements from initial prototyping

Next Steps: Testing with SLFs

- Replace hand crank with motor
- Real world testing needed
- Low cost and minimal assembly time make further expoloration logical